

PVsyst - Simulation report

Grid-Connected System

Project: caso4-campinas

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 1800 Wp

Campinas - Brazil



PVsyst V8.1.5

VCO, Simulation date:
09/09/26 14:04
with V8.1.5

Project summary

Geographical Site

Campinas
Brazil

Situation

Latitude -22.9056 °(S)
Longitude -47.0596 °(W)
Altitude 685 m
Time zone UTC-3

Project settings

Albedo 0.20

Weather data

Campinas
NASA POWER satellite data 2001-2020 - Synthetic

System summary

Grid-Connected System

Orientation #1 Horizontal plane

System information

PV Array

Nb. of modules
Pnom total

No 3D scene defined, no shadings

Near Shadings

no Shadings

User's needs

Unlimited load (grid)

Inverters

Nb. of units 1 unit
Total power 1.5 kWac
Pnom ratio 1.20

3 units
1800 Wp

Results summary

Simulation step Hourly

Produced Energy 2901.8 kWh/year Specific production 1612 kWh/kWp/year Perf. Ratio PR 85.93 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

Simulation step

Hourly

Orientation #1

Horizontal plane

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

no Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer Generic
Model JKM-600N-72HL4-V
(Original PVsyst database)
Unit Nom. Power 600 Wp
Number of PV modules 3 units
Nominal (STC) 1800 Wp
Modules 1 unit x 3 In series
At operating cond. (50°C)
Pmpp 1670 Wp
U mpp 122 V
I mpp 14 A

Total PV power

Nominal (STC) 1.80 kWp
Total 3 modules
Module area 7.7 m²
Cell area 7.2 m²

Inverter

Manufacturer Generic
Model MIC 1500TL-X
(Original PVsyst database)
Unit Nom. Power 1.50 kWac
Number of inverters 1 unit
Total power 1.5 kWac
Operating voltage 50-500 V
Pnom ratio (DC:AC) 1.20

Total inverter power

Total power 1.5 kWac
Number of inverters 1 unit
Pnom ratio 1.20

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 147 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.75 %

Module mismatch losses

Loss Fraction 1.00 % at MPP

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000



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Main results

Simulation step

Hourly

System Production

Produced Energy

2901.8 kWh/year

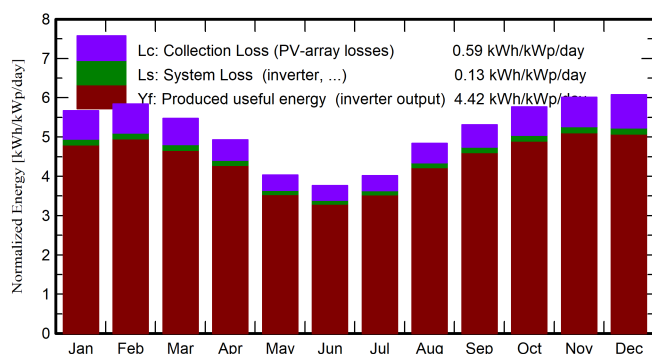
Specific production

1612 kWh/kWp/year

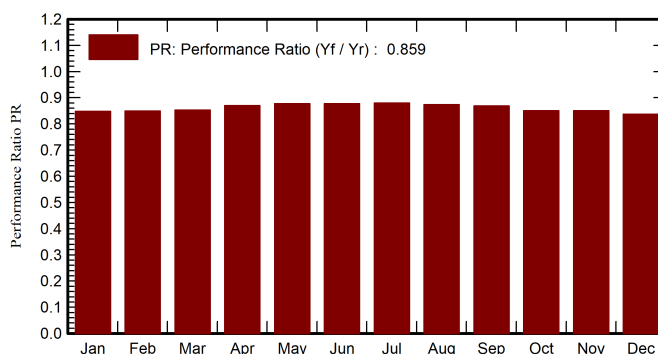
Perf. Ratio PR

85.93 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	175.8	83.60	22.84	175.7	171.5	276.4	268.2	0.848
February	163.6	66.87	22.90	163.5	159.5	257.5	250.0	0.849
March	169.7	64.91	22.14	169.7	165.1	268.5	260.6	0.853
April	147.9	47.79	20.65	147.9	143.3	238.5	231.5	0.870
May	125.2	41.71	17.76	125.1	120.6	203.7	197.7	0.878
June	112.9	33.95	17.03	112.8	108.0	183.6	178.2	0.878
July	124.7	34.13	16.80	124.6	119.5	203.2	197.4	0.880
August	150.0	36.87	18.37	150.0	145.2	242.9	235.8	0.874
September	159.3	49.12	20.66	159.2	155.2	256.3	248.9	0.868
October	178.9	69.89	22.20	178.8	174.4	282.0	273.9	0.851
November	180.4	78.40	22.07	180.4	176.0	284.5	276.1	0.850
December	188.3	85.91	22.83	188.3	183.8	292.1	283.5	0.836
Year	1876.7	693.16	20.51	1876.0	1822.1	2989.1	2901.8	0.859

Legends

GlobHor Global horizontal irradiation
DiffHor Horizontal diffuse irradiation
T_Amb Ambient Temperature
GlobInc Global incident in coll. plane
GlobEff Effective Global, corr. for IAM and shadings

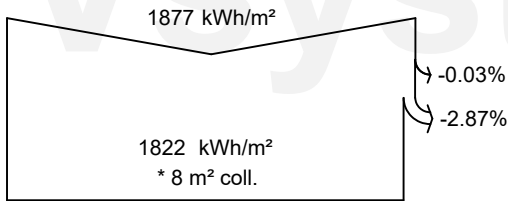
EArray Effective energy at the output of the array
E_Grid Energy injected into grid
PR Performance Ratio



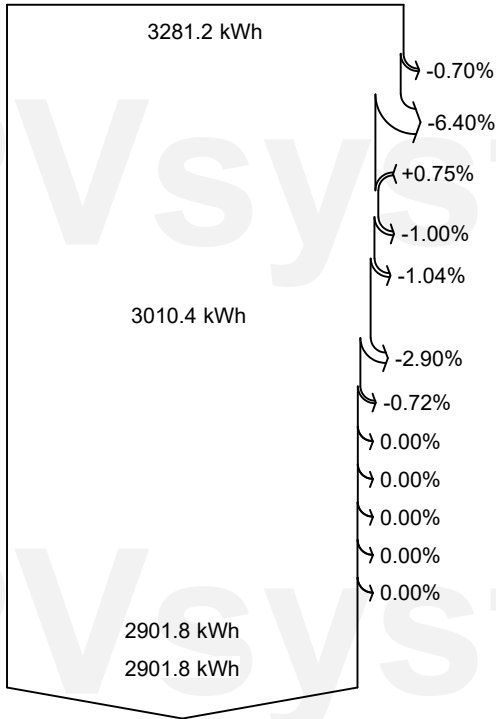
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Loss diagram



- Global horizontal irradiation
- Global incident in coll. plane
- IAM factor on global
- Effective irradiation on collectors
- PV conversion
- Array nominal energy (at STC effic.)
- PV loss due to irradiance level
- PV loss due to temperature
- Module quality loss
- Module array mismatch loss
- Ohmic wiring loss
- Array virtual energy at MPP
- Inverter Loss during operation (efficiency)
- Inverter Loss over nominal inv. power
- Inverter Loss due to max. input current
- Inverter Loss over nominal inv. voltage
- Inverter Loss due to power threshold
- Inverter Loss due to voltage threshold
- Night consumption
- Available Energy at Inverter Output
- Energy injected into grid



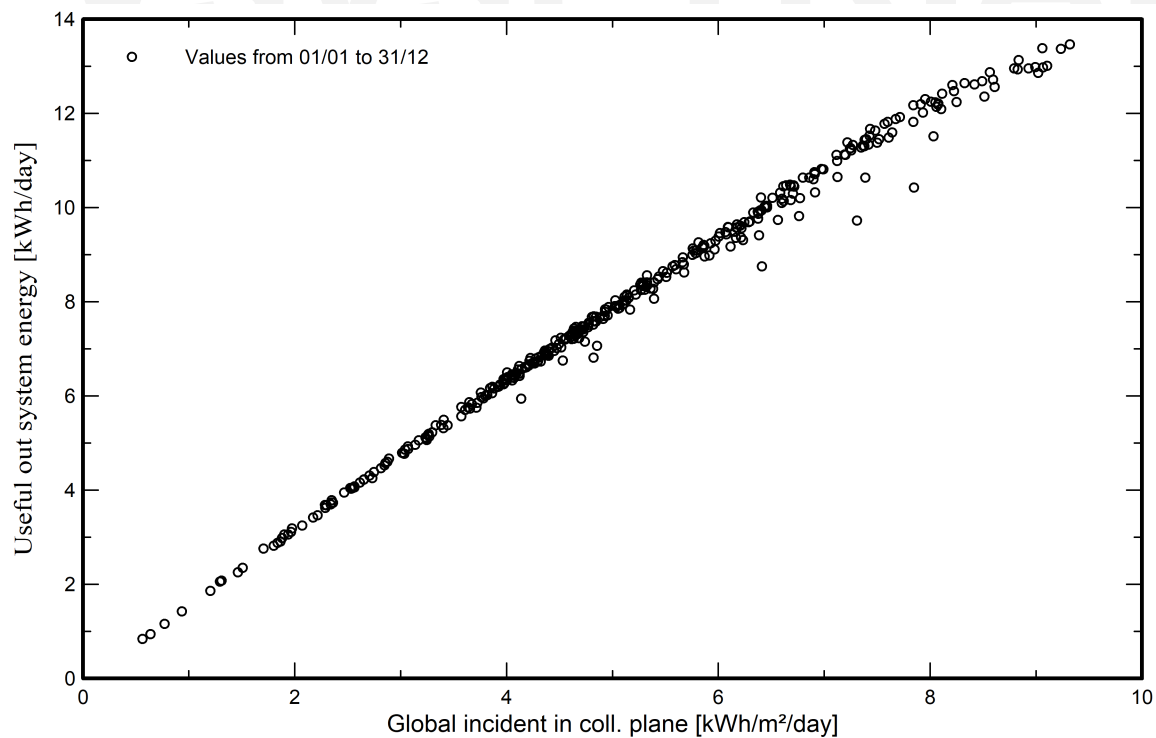


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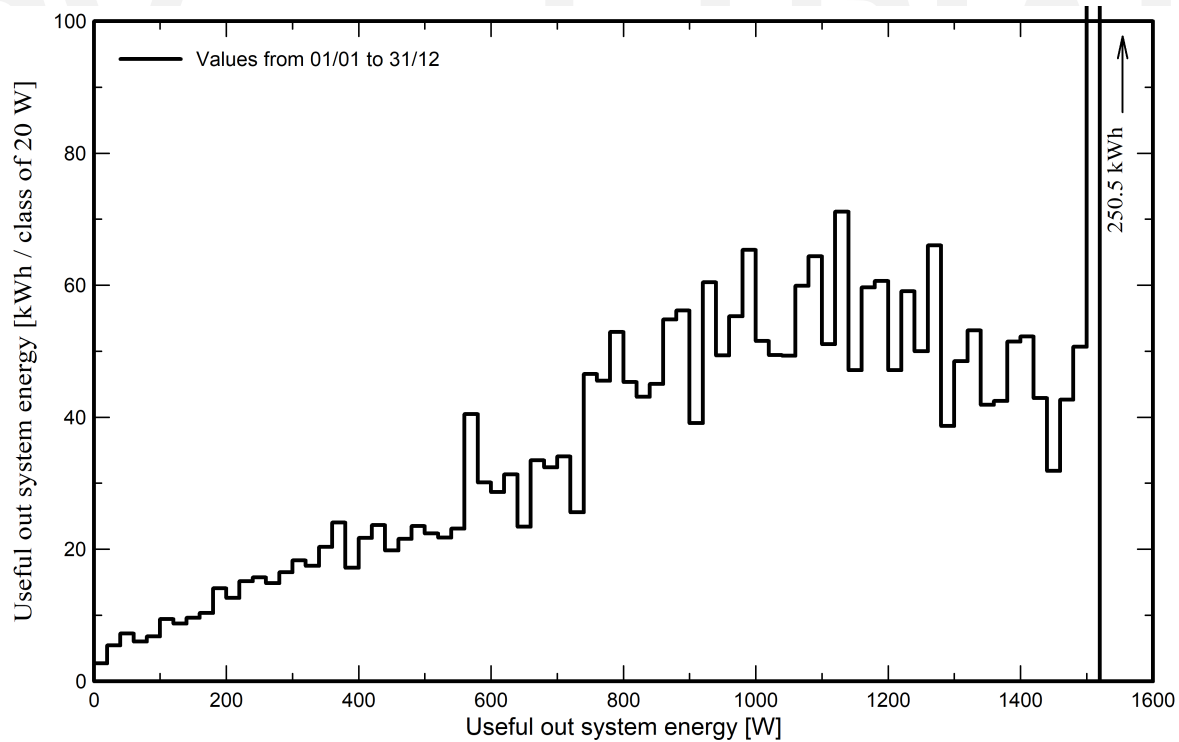
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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

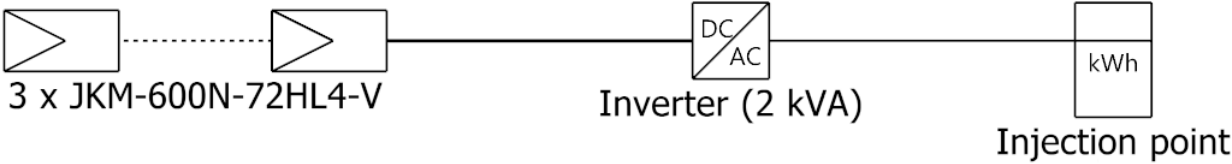




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Single-line diagram



PV module	JKM-600N-72HL4-V
Inverter	MIC 1500TL-X
String	3 x JKM-600N-72HL4-V

caso4-campinas

VC0 : New simulation variant

09/09/26